# Hardware–Algorithm Compatibility Report

## 1. Introduction

This report evaluates the compatibility between the implemented AEB algorithm and the configured sensor/control architecture in MATLAB/Simulink and IPG CarMaker.

## 2. Hardware Components

| Component | Function |
|---|---|
| Radar Sensor | Distance and relative velocity estimation |
| Stereo Camera | Object classification and validation |
| Vehicle Controller | Brake command execution |
| Simulink Environment | Control algorithm execution |
| IPG CarMaker | Vehicle dynamics simulation |

## 3. Algorithm Requirements

| Requirement | Supported By |
|---|---|
| Distance estimation | Radar |
| Relative velocity estimation | Radar |
| Object validation | Camera |
| Smooth brake generation | Simulink control blocks |
| Vehicle actuation | Vehicle control interface |

## 4. Compatibility Assessment

| Subsystem | Compatibility Status | Remarks |
|---|---|---|
| Radar + TTC Logic | Compatible | Reliable for collision estimation |
| Camera + Validation | Compatible | Suitable for object confirmation |
| Brake Interface | Compatible | Supports smooth braking |
| Simulink Logic | Compatible | Stable real-time operation |

## 5. Real-Time Stability

The implemented algorithm operates continuously with stable sensor sampling and low computational complexity. Transfer function smoothing and saturation logic improve overall stability.

## 6. Identified Limitations

| Limitation | Impact |
|---|---|
| Camera confidence variation | Reduced reliability at very close distances |
| Pixel threshold dependency | Scenario-specific tuning required |
| Ground truth simplification | Real-world noise not modeled |

## 7. Recommended Improvements

| Improvement | Expected Benefit |
|---|---|
| PID Brake Controller | Improved braking smoothness |
| Adaptive TTC Logic | Dynamic braking response |
| Kalman Filtering | Improved sensor fusion |
| Multi-object Tracking | Enhanced robustness |

## 8. Conclusion

The implemented AEB algorithm demonstrates strong compatibility with the selected hardware and simulation architecture, enabling reliable collision detection and smooth braking operation.